$$((x-1)+1)^2 = (x-1)^2 + 2(x-1) + 1$$
$$(x-1)^2 = (x-2)^2 + 2(x-2) + 1$$
$$x^2 = \sum_{b=0}^{x-1} [2b+1]$$
$$((x-1)+1)^1 = (x-1)^1 + 1 \; ((x-1))^1 = (x-2)^1 + 1$$
$$x^1 = \sum_{b=0}^{x-1} [1]$$
$$x^2 = \sum_{b=0}^{x-1} \left[2 \sum_{c=0}^{b-1} [1] + 1\right]$$
$$x^2 = 2 \sum_{b=0}^{x-1} \left[\sum_{c=0}^{b-1} [1]\right] + 1 \sum_{b=0}^{x-1} [1]$$

I have a theory, all x^n's can be written as a sum of nested sums with falling upper bounds

$$x^n = d_1 \sum_{b=0}^{x-1} 1 + d_2 \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} 1 + ... + d_n \overbrace{\sum_{b=0}^{x-1} \sum_{c=0}^{b-1} ... \sum_{n_n=0}^{n_{n-1}-1}}^{n \text{ sums}} 1$$

This the goal of this paper

Identities used

binomial therom
$(x+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b a^{r-b}\right]$
pascals identity
$\binom{a}{n} = \binom{a-1}{n} + \binom{a-1}{n-1}$

Reucrsions from Binomial therom

Case 1 – Non falling sums

$$(x+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b a^{r-b}\right]$$

let $a = 1$

$$(x+1)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b\right]$$

let $x = x' - 1$

$$((x'-1)+1)^r = x^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x'-1)^b\right]$$

in general

$$((x-b)+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x-b)^b a^{r-b}\right]$$

if b=a

$$x^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x-a)^b a^{r-b}\right]$$

we can subtitute the defintion of x^r into $(x-a)^b$ or a^{r-b}

$(x-a)^b$ is a good substitution as it the nested power term will shrink by a after $\frac{x}{a}$ expansions the inner power term will be 0

$$x^r = \sum_{b=0}^{r}\left[\binom{r}{b}(x-a)^b a^{r-b}\right]$$

$$(x-a)^b = \sum_{c=0}^{b}\left[\binom{b}{c}((x-a)-a)^c a^{b-c}\right]$$

$$x^r = \sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}((x-a)-a)^c a^{b-c}\right]a^{r-b}\right]$$

every layer strips one a from inner power's base

$$x^r = \sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}((x-a)-a)^c a^{b-c}\right]a^{r-b}\right]$$

$$x^r = \overbrace{\sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}\sum_{d=0}^{c}\left[\binom{c}{d}\cdots\left[\sum_{n_{\frac{x}{a}}=0}^{n_{(\frac{x}{a})-1}}\left[\binom{n_{(\frac{x}{a})-1}}{n_{\frac{x}{a}}}0^{n_{\frac{x}{a}}}a^{n_{(\frac{x}{a})-1}-n_{\frac{x}{a}}}\right]\right]a^{c-d}\right]a^{b-c}\right]a^{r-b}\right]}^{\frac{x}{a}\text{ sums}}$$

you can stop at any layer you want, you dont need to fully expand

$$x^r = \overbrace{\sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}\sum_{d=0}^{c}\left[\binom{c}{d}\cdots\left[\sum_{n_l=0}^{n_{l-1}}\left[\binom{n_{l-1}}{n_l}(x-la)^{n_l}a^{n_{l-1}-n_l}\right]\right]a^{c-d}\right]a^{b-c}\right]a^{r-b}\right]}^{l\text{ sums}}$$